

A Comparative Analysis for Loan Approval Prediction using Machine Learning

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Abstract—The financial sector of any nation is the key factor in elevating the economy of nation. The biggest issue that finances departments are facing is to approve loans for valid candidates only. Our proposed model for predicting the loan approval for valid candidates uses three different supervised learning algorithms for classification of loan approved or not, on the basis of candidate attribute values. In proposed study ensemble technique has been used to find out best classifier algorithm to train our model. In this paper we are comparing KNN, Decision Tree, Logistic Regression, SVC, and Naïve Bayes classifier algorithms based on their accuracy for provided data set to find out the highly accurate and efficient algorithm for designing a model for approving loan to valid candidate which will also help in minimizing the financial frauds.

Keywords—Machine learning, KNN, Decision Tree, Gaussian Naïve Bayse, Loan Approval, Fraud Prediction etc.

I. INTRODUCTION

In the era of automation, different models are been designed by researchers in every domain to reduce the human

efforts and increase the productivity of human beings. So far ‘n’ number of studies have been proposed on automated tasks in different disciplines of computing science. This proposed study is mainly focused on finding the highly accurate classifier algorithm to train our model for loan approval prediction. Many bank are facing issues of fraud loan application approvals. As ‘n’ number of applicants are applying for financial loan daily, so it becomes very tedious job for bank employees or finance employees to verify every application manually. So to reduce their work and to make the loan approval process more efficient and fast to handle the increased loan application, automated systems are required in finance departments. These automated systems could be designed using machine learning algorithms to train the model with labelled dataset.

As shown in figure 1, the banks or finance branches are receiving large amounts of applications for loan approvement by customers. But if employees of banks are going to review all those applications manually it will going to be very time

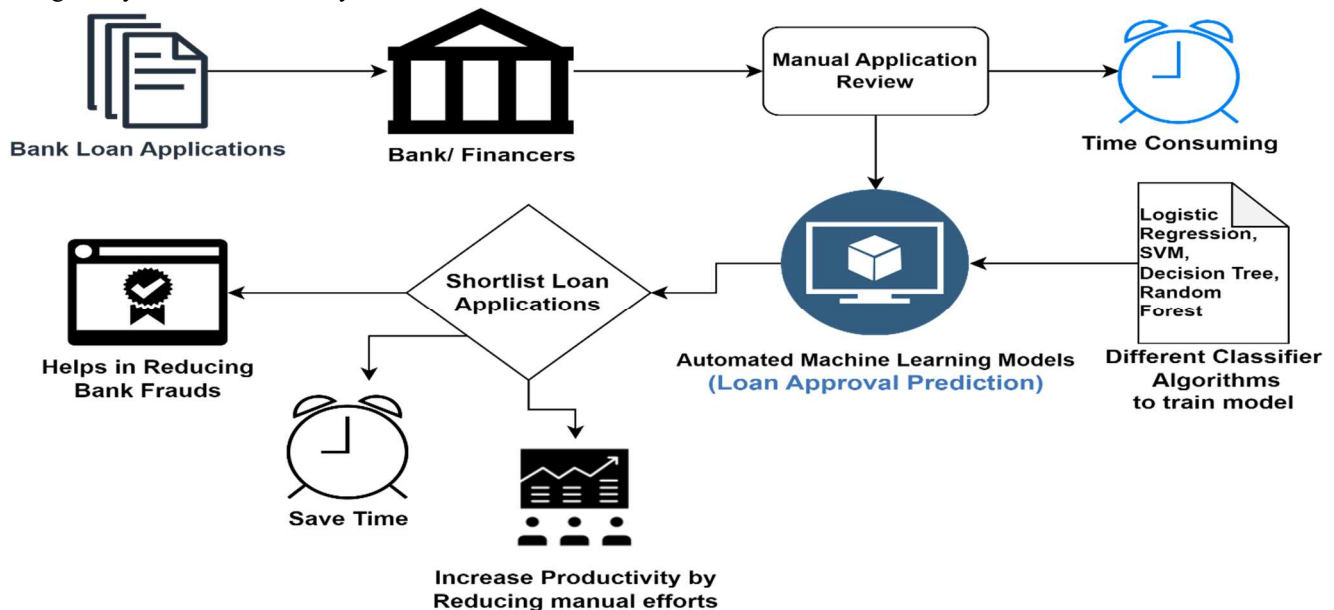


Fig. 1. Loan Approval Prediction Model Applications

consuming tasks as it will also affect the productivity and many fraud applications are also get accepted by some corruption or in illegal manner. To overcome these issues automated loan approval models could be build which offer a lot of benefit such as reduce the time consumption, it can also help in reducing the bank frauds and most import is it will definitely decrease the manual reviewing efforts of bank employees as some valid applications are going to be shortlisted by model and fraud applications or invalid applications are going to be rejected by model itself. As it reduces the time consumption, provide reliability to finance system, minimize the efforts which will definitely increase the productivity.

If we talk about the technicality of such model then it should be trained in such a manner that it can handle abnormalities in data also. The things or points that need to be taken care are model should be trained with large dataset, the most optimised classifier algorithm should be used to train the model to get highly accurate result or use ensemble techniques. In our model we are working with ensemble technique which consist different classifier algorithms and our aim is to compare the accuracy of all algorithms to find out the best one. The algorithm we have implemented on our data for classification are Decision Tree, Gaussian Naïve Bayse, Random forest, KNeighbors classifier and Support Vector Machine (SVM).

All the classifier algorithms have unique properties depending on the problem statement. If we talk about decision tree classifier algorithm it handles both categorical as well as numerical data and no need for feature scaling in most cases. It is easy to understand and interpret as we can visualize data in tree's structure. The Gaussian Naïve Bayes classifies calculates the data point probability belonging to each class based on these independent probabilities which is efficient for large datasets but main point is it works well with categorical features. The random forest is an ensemble method which combines multiple decision trees. It is highly accurate and robust to noise due to ensemble learning and handles high-dimensional data very well and it is also less prone to overfitting than individual decision tree classifier. The KNN classifies the data point based on the majority class of its k nearest neighbors in the training data. The support vector machine (SVM) aims to find hyperplane in feature space which helps to best separates the data points of different classes with maximum margin. So SVM handle non-linear data and good for small datasets and noisy data.

Overall, for designing a loan prediction model training is the most important. The ensemble technique offers better results for training models with test data and the general point of view is that choosing the classifier algorithm is totally dependent on the type of dataset we are using to train our model as each algorithm have their unique qualities and effectiveness on model and limitations as well.

II. RELATED WORK

To propose this study or research paper, we have gone through enormous number of literature or research papers related to our topic of research. We analysed and compared research work of different researchers to understand how research is going on for loan approval perdition purposes. N.Boutaher et al. [1] proposed review study on frauds detection of credit card which reviewed different supervised learning techniques such as logistic regression, random-forest

and SVM. Then M. Alaradi et al. [2] proposed study on “tree based methods for loan approval” which shown results for decision tree, bagging, random forest, boosting but data description such as type of data or data size is missing.

TABLE I. LITERATURE REVIEW

Author (Year)	Methodology / Algorithm	Accuracy	Research Gap
Ramchandra H. V. et al. [3] (2021)	Decision Tree, Random Forest	86%	Doesn't particularly defined accuracy for different techniques. Dataset details missing.
Vishal Singh et al. [4] (2021)	XGBoost, Random Forest, Decision Tree	Good	Accuracy not mentioned in numerical value.
Praveen Tumuluru et al. [5] (2022)	SVM, KNN, Random Forest, Logistic Regression	73.2%, 68%, 81%, 77% respectively	Dataset size is missing in the form of rows and columns.
Ch. Naveen Kumar et al. [6] (2022)	Decision Tree, Random forest, SVM, KNN	69%, 72%, 70%, 59%, respectively	Huge difference in train and test data accuracy.
B.P. Lohani et al. [7] (2022)	Logistic Regression	Not mentioned	Model accuracy not mentioned.
U.E. Orji et al. [8] (2022)	Logistic Regression, KNN, SVM, Decision Tree, Bagging and Boosting	Highest 95.55% for Random forest, Lowest 80% for logistic regression.	Doesn't classified accuracy results for training and testing data.
Nancy D.R. et al. [9] (2023)	KNN, Decision Tree, Support Vector Classifier	83% for SVM	More attributes should be considered in dataset.
P. Nagaraj et al. [10] (2023)	Logistic Regression, KNN, Random Forest, Extra tree classifier	82.1%, 77.3%, 87.03%, 82.70% respectively	Dataset description missing.

H.K. Sarisa et al. [11] (2023)	Logistic Regression, Decision tree, Random forest.	RF-76.66%, DT-67.52%, KNN-80.83%	Abstract statement not matched with results.
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The above table represents the comparison of studies proposed by different researchers for loan approval prediction. With the increasing number of loan applications in banks it is tough and risky as well to check each person's application individually [12] by bank employees. So researchers are working on developing loan prediction systems to shortlist the valid candidate to whom loan should be given or not.

So overall researchers have applied enormous numbers of supervised learning algorithms to achieve the highest accuracy for loan approval models. The above literature review table also shows that data set plays major role in training our model. So our literature review shows that the most common algorithms that researchers have used so far are decision tree, random forest, KNN and SVM.

III. METHODOLOGY

The implementation of this proposed work is done using Python programming language and we have used Jupyter Notebook for implementing our designed model to verify the accuracy of our model with different learning algorithms to train our model. The input to model is dataset which involves 615 rows data and attributes as shown in table 2. The output from our model is the decision predicted by our model that customer will get approval for loan or not. In next section we are going to discuss about our data set and preprocessing steps.

A. Dataset Description

The dataset we are using in our model is available on Kaggle [13] "Loan Prediction". The size of the dataset is 615 rows x 13 columns. So it means that dataset consists of 13 unique features which will train our model. The following table gives the brief description about our dataset attributes.

TABLE II. DATASET ATTRIBUTES

Variable	Description	Data Type
Loan_ID	Unique Application Number	Numeric Data
Gender	Specify Male/Female	Categorical Data
Married	Marital Status	Categorical Data
Dependents	No. of family members	Numeric Data
Education	Qualificaion in terms of study	Categorical Data
Self_Employed	Employment status	Categorical Data
ApplicantIncome	Salary of applicant	Numeric Data
CoapplicantIncome	Additonal applicant's salary	Numeric Data

LoanAmount	Total loan amount	Numeric Data
Loan_Amount_Term	Loan's Return Period	Numeric Data
Credit_History	Record for applicant's credit history	Numeric Data
Property_Area	Location or home (Rural/Urban)	Categorical Data
Loan_Status	Approval Status of loan	Categorical Data

B. Data Preporcessing and Data Analysis

In preprocessing of collected data various steps are involved to remove the noise or impurity of our dataset. We listing three most important steps below which falls under data preprocessing, many other steps are also there but these are major ones.

1) *Handling Missing Values*: The data may consist of some empty cells so we have fill that cell in our implementation we have used fillna() method to fill the missing values.

2) *Handling Categorical Variable*: As we know machine learning algorithms typically works with numerical data so we need to convert our categorical data into numbers and one of the common method to do the same is one-hot encoding which we have used in our model.

3) *Splitting Data*: Important step to before we start training our model, we need to split our data into two categories i.e., training set and test set. The training set we use to teach our model under supervised learning and test set we use to check the performance of our model that how accurately it is working.

Furthermore, figure 2 represents the correlation of all the attributes of our dataset with each other which shows that Applicant Income, Loan Amount, and Credit History plays major role in approving the loan or rejecting the loan application.

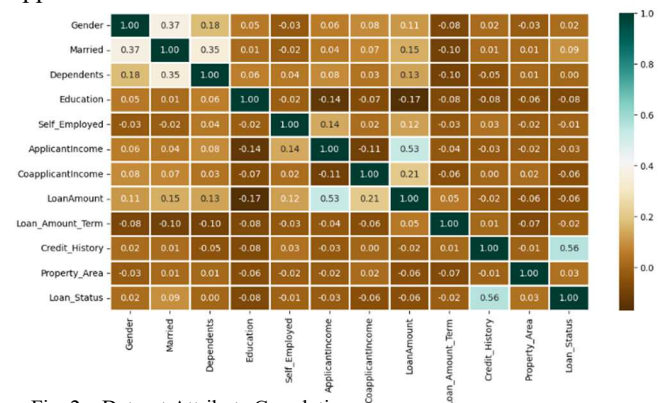


Fig. 2. Dataset Attribute Correlation

The following figure 3, represents the loan approval relation between categorical data points of our data set. The figure represents the categorical data points relationship in bar graph from which is easy to understand and analyse.

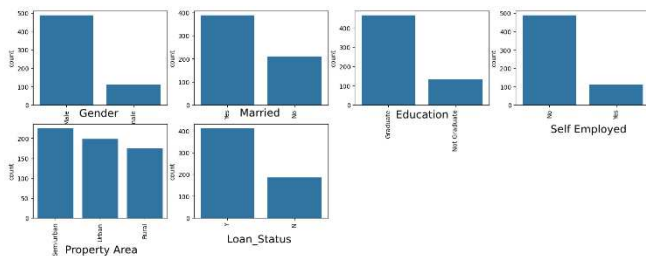


Fig. 3. Categorical Data Correlation

So the above figure 3 represents the total 6 categorical data graphs which shows that there are high chances of get loan approval if person is male, married, graduate, self-employed and living in simulation or urban areas. The implementation process of our model is discussed in next section with accuracy of different machine learning algorithms.

IV. IMPLEMENTATION

In our model we have implemented different supervised learning classifier algorithms and we find out the best algorithm based on the accuracy offered by our model in both training and testing phases.

First step was to import all the necessary libraries and data set to our Jupyter notebook as shown in figure 4.

Once the data is loaded successfully to our project, we checked for categorical data and converted it to numeric values using one hot encoder. Then next step was to remove

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History	Property
0	LP001002	Male	No	0.0	Graduate	No	5849	0.0	NaN	360.0	1.0	
1	LP001003	Male	Yes	1.0	Graduate	No	4583	1508.0	128.0	360.0	1.0	
2	LP001005	Male	Yes	0.0	Graduate	Yes	3000	0.0	66.0	360.0	1.0	
3	LP001006	Male	Yes	0.0	Not Graduate	No	2583	2358.0	120.0	360.0	1.0	
4	LP001008	Male	No	0.0	Graduate	No	6000	0.0	141.0	360.0	1.0	

Fig. 4. Dataset Outlook

data impurities such as fill the missing values we used fillna() method to do the same which fill the missing values by taking the mean of other values. After this we checked our dataset to confirm that data is in pure form and ready to train our model. The figure 5 is attached to show that how pure data output looks like.

```
for col in data.columns:
    data[col] = data[col].fillna(data[col].mean())

data.isna().sum()

Gender                0
Married               0
Dependents            0
Education             0
Self_Employed        0
ApplicantIncome       0
CoapplicantIncome     0
LoanAmount            0
Loan_Amount_Term      0
Credit_History       0
Property_Area         0
Loan_Status           0
dtype: int64
```

Fig. 5. Filling all the missing values

Then we divided our dataset into two sets, one is training set and other is testing set. We implemented ensemble techniques which means more than one algorithms in simple words we have used total 6 algorithms to train our loan prediction model and these algorithms are Random Forest Classifier, K-Neighbors Classifier, Support Vector Classifier, Logistic Regression, Decision Tree Classifier and Gaussian

Accuracy score of RandomForestClassifier = 98.04469273743017
 Accuracy score of KNeighborsClassifier = 78.49162011173185
 Accuracy score of SVC = 68.71508379888269
 Accuracy score of LogisticRegression = 79.60893854748603
 Accuracy score of DecisionTreeClassifier = 69.16749999999999
 Accuracy score of GaussianNB = 79.05027932960894

Fig. 6. Training Data Loan Prediction Model Accuracy

Naïve Bayes Classifier. The figure 6 and figure 7 represent the accuracy output of our model in both training and testing phase respectively.

The figure 6 represents the accuracy output of algorithms applied on training data set which shows random forest algorithm is giving highest accuracy while figure 7 represents the accuracy output of algorithms performed on testing data

Accuracy score of RandomForestClassifier = 82.5
 Accuracy score of KNeighborsClassifier = 63.74999999999999
 Accuracy score of SVC = 69.16666666666667
 Accuracy score of LogisticRegression = 80.83333333333333
 Accuracy score of DecisionTreeClassifier = 68.33333333333333
 Accuracy score of GaussianNB = 81.66666666666667

Fig. 7. Testing Data Loan Prediction Model Accuracy

set and accuracy slightly decreased than training phase.

The above figures are showing the results of our implemented model and our model performed better than other models listed literature review as 3 algorithms out of 6 are giving accuracy above 80%. For better comparison refer to Table 3.

TABLE III. MODEL ACCURACY ON TRAINING AND TEST DATA

Sr. No.	Algorithm	Accuracy (Training)	Accuracy (Test Data)
1	Random Forest Classifier	98.04%	82.5%
2	KNeighbors Classifier	78.49%	63.74%
3	Support Vector Classifier	68.71%	69.16%
4	Logistic Regression	79.60%	80.83%
5	Decision Tree Classifier	69.16%	68.33%
6	Gaussian Naïve Bayes	79.05%	81.66%

The above table 3 represents the model accuracy results for our model on training data and test data. The random forest classifier is giving highest accuracy i.e., 98.04% while Support Vector Classifier is giving lowest accuracy i.e. 68.71% in training phase. While in test phase the random forest algorithm's accuracy decreases but still highest as compared to other algorithms. Overall comparison says that Random forest, logistic regression, and Gaussian Naïve Bayes

algorithms are constantly giving high accuracy result in both training as well as testing phase.

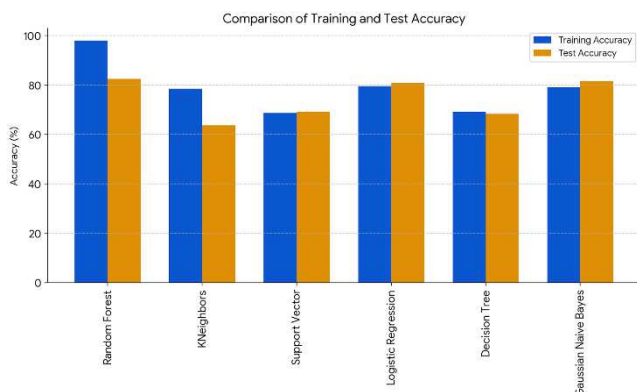


Fig. 8. Graphical Representation of Accuracy

The above figure 8, represents the graphical representation of comparison all the algorithms we have used to train our loan prediction model. The graph shows that Random Forest is giving highest accuracy results to our model in both training and testing phase also. In testing phase, Random Forest, Logistic regression and naïve bayes are giving quite similar results which also represents the quality of dataset we have used in this research.

V. RESULT

The study proposed a comparison of different supervised learning models for loan approval prediction model implemented by us. The results shown in table 3 shows that Random Forest is the best algorithm on our dataset as it is giving 98.04% accuracy during training phase and 82.5% accuracy in testing phase. The accuracy of decision tree and support vector classifier comes out to be very low, in training phase 68.71% and 69.16% respectively while in testing phase it comes out to be 69.16% and 69.33% respectively. Rest all models are giving almost similar accuracy result in testing phase of our model. Overall random forest stands out to be best from all the six algorithms we used implemented.

VI. CONCLUSION

Though we have performed all the steps of machine learning from importing the data to visualization of the data through univariate and bivariate analysis, and found their importance on the final column, which is the result of our dataset, loan status where banks can significantly calculate with the help of the previous data of the person whether the bank has provided the loan for a given person or not, in yes or no format. Though in our research analysis, it is showing that random forest outperforms all the other algorithms. But in the future, we can improve our accuracy with the help of deep learning algorithms and ensemble algorithms for machine learning. After that, further, we can improve the frontend and backend with the help of mobile applications or the web application, which is easy to use.

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